

ANUJ KUMAR

Formal Verification Engineer

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SUMMARY

Electronics & Communication Engineering student with hands-on ASIC and SoC front-end formal verification experience: writing SystemVerilog RTL and SystemVerilog Assertions (SVA), and proving properties with Cadence JasperGold and Synopsys VC Formal on a RISC-V pipeline, a GPU MAC datapath, and a MESI cache-coherence protocol. Comfortable reading RTL quickly and reasoning about hazards, liveness, and safety with temporal logic. Background also includes bare-metal embedded C and systems programming, bringing a strong grasp of hardware/software interaction to verification work.

TECHNICAL SKILLS

Formal Verification & Digital Design: SystemVerilog (RTL & SVA), Verilog, Formal Property Verification, Cadence JasperGold, Synopsys VC Formal, Bounded Model Checking (BMC), K-Induction, State-Space Reduction, Verilator (lint/simulation)

Computer Architecture: RISC-V (RV32I) pipelined datapath design, hazard detection & forwarding, cache coherence protocols (MESI), bus arbitration, memory systems

Languages & Scripting: C/C++, Embedded C, Python, TCL, SQL, Data Structures & Algorithms, Object-Oriented Design

Embedded & Tools: ARM Cortex-M, ARM CMSIS, RTOS, UART/I2C/SPI, Git, Docker, GDB, Valgrind, CI/CD, Linux

EDUCATION

JSS Academy of Technical Education — B.Tech, Electronics & Communication Engineering (Nov 2022 – Jul 2026)

Noida, India | Class XII (CBSE): 94.6%

- Relevant Coursework: Embedded Systems, Microprocessors, Computer Architecture, Operating Systems, Digital Signal Processing (DSP), System Design

PROJECTS — FORMAL VERIFICATION & DIGITAL DESIGN

Formal Verification of a RISC-V Microprocessor (RV32I) | *SystemVerilog, SVA, Cadence JasperGold, Verilator, Python*

- Designed and verified a custom 5-stage pipelined RV32I core (IF/ID/EX/MEM/WB) with operand forwarding, hazard detection, and branch misprediction flush recovery.
- Authored 50+ concurrent SVA properties covering pipeline hazards, forwarding correctness, register-file consistency, and deadlock-freedom (liveness); proved properties in Cadence JasperGold.
- Built a Python pipeline to parse and triage formal proof reports, and used Verilator for static lint and compilation checks in a local CI flow.

GPU Tensor Core MAC Unit — Formal Verification | *SystemVerilog, Synopsys VC Formal / JasperGold, Python*

- Designed a parameterized 3-stage signed fixed-point multiply-accumulate (MAC) unit with saturating overflow checks and a ready-valid handshake control FSM.
- Built the formal verification environment and a Python generator producing 100+ bit-sliced SVA properties, localizing checks so formal engines converged in under one minute.
- Used cut-point/symbolic abstraction to black-box the multiplier stage, avoiding state-space explosion on a 2^{32} input space while proving downstream accumulator saturation bounds.

MESI Cache Coherence Protocol (GPU L2) | *SystemVerilog, SVA, Cadence JasperGold*

- Implemented synthesizable RTL for a dual-core cache cluster with snooping controllers, a round-robin bus arbiter, and a memory controller.
- Formally verified 19 concurrent SVA properties: mutual exclusion between Modified/Exclusive and Invalid states, shared-data consistency across cores, snoop-invalidation invariants, and arbiter starvation-freedom.

ARM Cortex-M3 UART-I2C Protocol Bridge | *Embedded C, ARM CMSIS, STM32F103*

- Built interrupt-driven UART-to-I2C bridge firmware using a lock-free SPSC ring buffer, with hardware data memory barriers (`__DMB()`) to prevent reordering.
- Wrote a host-based register-mocking test harness enabling native unit testing of drivers and protocol parsing without physical hardware; offloaded bulk transfers to DMA, cutting ISR latency 70%.

EXPERIENCE

Software Development Intern — Quizzoc (Remote) (Jan 2024 – Mar 2024)

- Cut page load time by 10%, confirmed through frontend performance testing, by implementing lazy loading and optimizing API calls across full-stack features.
- Shipped secure REST APIs supporting 50+ monthly active users, validated through Postman testing, by applying clean API design with integrated authentication.

CERTIFICATIONS & ACHIEVEMENTS

- Data Structures & Algorithms: 500+ algorithmic problems solved across LeetCode, CodeChef, and GeeksforGeeks, optimizing for time and space complexity.
- AWS Certified Cloud Practitioner — Amazon Web Services | FreeCodeCamp (5x) | Forage Job Simulations (4): Citi, Commonwealth Bank, Skyscanner, Walmart Global Tech.